Ontologies, Digital Twin and Intelligent Telemetry for Internet Infrastructure (DT-II)

Juliao Braga
Center for Mathematics, Computation and Cognition
Federal University of ABC
Santo André, SP, BR
juliao.braga@ufabc.edu.br

Jéferson Campos Nobre
Instituto de Informática - PPGC
Universidade Federal do Rio Grande do Sul
Porto Alegre, RS, BR
jcnobre@inf.ufrgs.br

Juliana C. Braga
Center for Mathematics, Computation and Cognition
Federal University of ABC
Santo André, SP, BR
juliana.braga@ufabc.edu.br

Itana Stiubiener
Center for Mathematics, Computation and Cognition
Federal University of ABC
Santo André, SP, BR
itana.stiubiener@ufabc.edu.br

Leandro Marcio Bertholdo
Instituto de Informática - PPGC
Universidade Federal do Rio Grande do Sul
Porto Alegre, RS, BR
leandro.bertholdo@gmail.com

Marcelo Santos
Instituto de Informática
IFSertão Pernambucano
Salgueiro, PE, BR
marcelo.santos@ifsertao-pe.edu.br

Percival Henriques
Comitê Gestor da Internet no Brasil
São Paulo, SP, BR
percival@cgi.br

Abstract

This project proposes a semantic unification for Internet Infrastructure through the use of a High-Level Ontology (Upper Ontology) grounded in the Unified Foundational Ontology (UFO) and expressed in the axioms of Description Logic (TBox, ABox, and RBox), based on the OWL (Web Ontology Language) of the Semantic Web invented by the W3C (World Wide Web Consortium). The approach employs modeling tools such as OntoUML, which facilitates the use of mereology and relators to establish a "Logical Constitution" that transcends specific technological implementations (such as Wireless Networks, Optical Fiber, or Satellites). The resources are organized into a fundamental triad composed of Network Node, Connection, and Channel. By integrating this rigorous model into a dynamic Knowledge Graph, ensuring Universal Semantic Interoperability and the use of Reasoners, the proposed ecosystem enables automated fault detection, cascading impact analysis, and complex governance management, including Service Level Agreements (SLA) and Semantic Jurisdiction. The result is a Digital Twin model that transforms passive infrastructure into an autonomic and resilient system, capable of providing operational intelligence and real-time decision support.

Keywords domain ontology · upper ontology · ufo · ontouml · netconf · digital twin

1 Introduction

The relationship between ***Ontology*** and the ***Knowledge Graph*** (KG) is central to how modern ***Artificial Intelligence*** (AI) organizes information Kejriwal et al. [2021], Landgrebe and Smith [2025], Braga et al. [2025]. In the context of Internet Infrastructure (II), Ontology provides the logical blueprint, while the KG represents its dynamic instantiation. This project proposes an ***Upper Ontology*** (UO) for Internet Infrastructure (II) Arp et al. [2015], structured into four pillars:

P1. **Theoretical Foundation**: Ontology defines classes, relations, and axioms, while the KG organizes facts into `Subject-Predicate-Object` triples. ***Description Logic*** (DL) Baader et al. [2017], Nardi and Brachman [2010], Horrocks et al. [2006] ensures computational rigor for hierarchies and constraints.

P2. **Modeling Methodology**: Using the ***OntoUML Lightweight Editor*** (OLED) Guerson et al. [2015], Guizzardi et al. [2018], the ontology distinguishes essence (*Kind*) from role (*Role*), exporting models into interoperable formats such as ***Turtle*** and ***OWL*** W3C OWL Working Group [2012].

P3. **Upper Ontology**: A **Logical Constitution** unifies subdomains (e.g., 5G, satellites, optical fibers) into the triad: **Node**, **Connection**, and **Channel**.

P4. **Operational Intelligence**: A ***Digital Twin*** Grieves and Vickers [2017], Zhou et al. [2026], operated by an ***Inference Engine***, integrates ***TBOX***, ***ABOX***, and ***RBOX*** Nardi and Brachman [2010], Horrocks et al. [2006] to detect failures and manage governance, including ***Service Level Agreements*** (SLA) and semantic jurisdictions[1].

The instantiation of the Digital Twin relies on resources associated with the NETCONF protocol, including YANG-Push, as well as complementary mechanisms such as gNMI, and, in the case of legacy systems, SNMP Claise et al. [2019], Enns et al. [2011], Bjorklund [October 2010].

Figure 1 synthesizes the project's **Intelligence Cascade**, showing how the UO, domain ontologies, and Digital Twin interact to ensure semantic interoperability and dynamic governance.

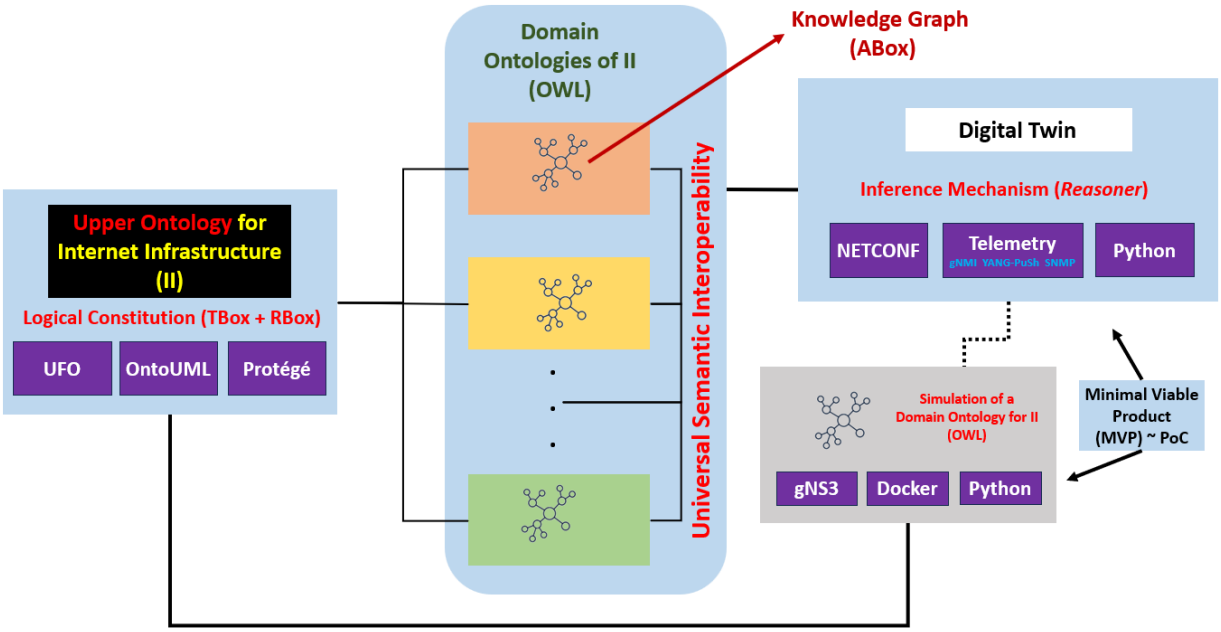

Figure 1: Overview of the DT-II Project Architecture

[1]Semantic Jurisdiction is understood as a normative domain of logical authority, formally established through ontological axioms, that regulates the behavior of digital entities based on meaning, context, and relations. Unlike geographic boundaries or physical attributes, it functions as a governance layer mediated by Relators, capable of propagating restrictions and norms (such as laws, service-level agreements, and privacy policies) across nodes, connections, and channels within an infrastructure. Its purpose is to ensure cross-domain compliance through automated reasoning, allowing rules to be verified, inherited, and dynamically applied to data flows and network operations, thereby safeguarding normative integrity in complex and evolving environments.

2 Methodology

This project follows the ontology engineering lifecycle Keet [2025] Parks [2025] Effingham [2013] Kendall and McGuinness [2019], adopting the ***Unified Foundational Ontology*** (UFO) as the basis and OntoUML as the modeling language Guizzardi [2005]. The process is structured into four technical phases:

M1. **Conceptual Abstraction**: UFO theory is applied to classify network assets. Rigid essence classes

$$(\texttt{PhysicalElement}, \texttt{MaterialMedium})$$

define identity, while accidental roles

$$(\texttt{NetworkNode}, \texttt{CommunicationChannel})$$

are linked through Relators.

M2. **Formalization in DL**: Concepts are axiomatized in ***Description Logic*** (DL), ensuring computational interpretability. For example, every `NetworkNode` must possess a logical identifier:

$$\text{NetworkNode} \sqsubseteq \text{PhysicalElement} \sqcap \exists \text{possesses.LogicalIdentifier} \tag{1}$$

M3. **Semantic Serialization**: Validated models in ***OLED*** are exported to OWL and serialized in ***Turtle***. Standard vocabularies such as ***Dublin Core*** (DC) and ***FOAF*** support metadata and provenance.

M4. **Automated Reasoning**: Inference engines confront the ***TBox*** and ***ABox*** to check consistency and detect anomalies. Predicate logic enables deduction of failure states, e.g., an inactive Channel implies compromise of its logical connection.

2.1 Ontological Analysis Framework and Mapping Process

The mapping between the II domain and the UFO through its implementation, ***gUFO***, is not arbitrary; it follows a specialization protocol grounded in logical metaproperties. For each entity in the local domain, the **Ontological Analysis Framework** is applied Almeida et al. [2026, 2019], Guizzardi [2005]:

A. **Specialization of Fundamental Types:** Entities with rigid identity and specific principles of individuation (e.g., Physical Routers, Satellites, Fiber Cables) are mapped as subclasses of *gufo:Kind*. These represent substantial objects that persist regardless of their configuration state.

B. **Relational Specialization (Roles):** Concepts whose existence depends on a specific topological or functional context (e.g., Border Node, Peering Point, Transit AS) are characterized as *gufo:Role*. This allows the Digital Twin to capture the dynamic nature of network functions independently of the underlying hardware.

C. **Relational Foundations:** Logical connections and service agreements that establish dependencies between multiple objects (e.g., BGP Sessions, SLA Agreements) are formalized as *gufo:Relator*. This ensures that the Knowledge Graph integrity is maintained through the existential dependence constraints inherent in gUFO, where the invalidation of a relator triggers the logical removal of the associated roles.

By employing this framework, the DT-II transcends simple data modeling, establishing a "Logical Constitution" where network behavior is governed by foundational ontological axioms rather than purely heuristic or statistical patterns.

2.2 Dynamic Synchronization via YANG-Push

A critical requirement for the DT-II is the high-fidelity synchronization between the physical infrastructure and its semantic representation. To achieve this, the model will employ *Model-Driven Telemetry* (MDT) through **YANG-Push** mechanisms (RFC 8641 Clemm and Voit [2019]).

Unlike traditional pull-based monitoring, YANG-Push allows the Network Nodes to stream state changes directly to the *Semantic Ingestor*. This integration is mapped onto the gUFO framework as follows:

- **Periodic Subscriptions:** Quantitative metrics, such as interface counters, are streamed to update `gufo:Quality` assertions in the ABox.
- **On-change Notifications:** Critical events, such as BGP session flaps, are treated as `gufo:Events` that immediately trigger the re-evaluation of existential dependencies in the `gufo:Relators`.

This ensures that the Digital Twin's "Logical Constitution" always reflects the current operational state of the ASx with sub-second latency.

3 Relations between Ontology, Knowledge Graph and Description Logic

The ***Knowledge Graph*** (KG) operationalizes ontology by linking **real data** (instances) to the conceptual model. It represents information as **nodes** (entities) connected by **edges** (relations). For example, the node `The Godfather` is linked to `Francis Ford Coppola` through `directed_by`. Typically, a KG integrates multiple sources to form a web of interconnected facts.

Ontology and KG are **complementary**: Ontology provides semantics and logical rules, while the KG instantiates them as a queryable and dynamic knowledge base. Table 1 summarizes their roles in the proposed framework.

Table 1: Complementarity between Ontology and Knowledge Graph

Characteristic	Ontology	Knowledge Graph
Focus	Definitions, rules, logic.	Data, facts, connections.
Level	Abstract (the "DNA").	Concrete (the "Organism").
Function	Provides meaning and semantics.	Enables querying and discovery.

Description Logic (DL) underpins modern ontologies (e.g., OWL in the Semantic Web) Baader et al. [2017]. DL enables formal concept definition and computational reasoning. For instance, the axiom

$$\text{Director} \sqsubseteq \text{Person}$$

states that every Director is a Person. More complex definitions use logical connectives, such as:

$$\text{Filmmaker} \equiv \text{Person} \sqcap \exists \text{directed.Film}$$

meaning that a Filmmaker is a Person who has directed at least one Film. Here, $\sqcap$ denotes intersection (AND), ensuring precise semantic constraints.

4 Conceptual Structure of the Digital Twin

This section presents a **reference architecture** for implementing the Digital Twin of Internet Infrastructure (II) as an **operational KG** (***ABox***) governed by an ***UO*** (***TBox***, ***RBox***). The objective is to make the model actionable: collect, normalize, validate, infer, and support decisions. The implementation objectives are: (a) **Unified semantic inventory**: represent nodes, connections, and channels independently of technology; (b) **Continuous synchronization**: align the KG with telemetry and topology changes; (c) **Detection and diagnosis**: identify inconsistencies, failures, and cascading impacts, and (d) **Verifiable governance**: encode and audit rules (SLA, jurisdiction, authorization).

4.1 Reference Architecture

The architecture is organized into seven layers: (i) **Ontological Layer (*TBox*)**: defines core classes (`NetworkNode`, `Connection`, `Channel`) and constraints; (ii) **Identity Layer**: assigns stable IRIs to instances, separating identity from state; (iii) **Ingestion Layer**: collects telemetry (YANG-Push, gNMI, SNMP, syslog) and normalizes data into the triad Node–Connection–Channel; (iv) **Persistence Layer (*ABox*, *RBox*)**: stores facts in RDF/OWL triplestores and maintains temporal history via observations; (v) **Validation Layer**: applies structural constraints and logical consistency checks with DL reasoners; (vi) **Reasoning Layer**: infers states (`DegradedConnection`, `SLAViolation`) and propagates failures; (vii) **Exposure Layer**: provides SPARQL queries, dashboards, alarms, and integration with NOC/orchestration systems.

4.2 Minimal Data Model

The initial implementation focuses on three entities: (a) **Node**: identity (`hasIdentifier`), operational state, timestamp; (b) **Connection**: terminals (`connectNode`), supported channel, metrics (latency, loss, jitter); (c) **Channel**: medium type (fiber, copper, radio), capacity, QoS policy.

4.3 Integration of *ABox* and *RBox*

The ***ABox*** instantiates infrastructure elements (routers, links, channels) from inventory data Claise et al. [2019]. The ***RBox***, based on $\mathcal{SROIQ}$ logic Horrocks et al. [2006], governs relational properties: (a) **Transitivity**: deduces

cascading failures; (b) **Mereology**: validates part–whole relations for SLA compliance [Effingham, 2013, Chap. 8], (c) **Property chains**: enable bidirectional navigation and anomaly tracing.

Together, ***ABox*** and ***RBox*** transform the KG into a computable model, enabling dynamic reasoning and governance.

4.4 Operational Flow

F1. Ingestion → normalization → triple generation.

F2. Validation → integrity checks and anomaly detection.

F3. Inference → classification and rule-based derivations.

F4. Query/alert → recommendations or automated orchestration.

4.5 Scenario Example

In a fiber link failure: (a) telemetry marks the `Channel` as unavailable; (b) inference classifies the `Connection` as `InactiveConnection` and derives `SLAViolation`; (c) redundancy is triggered (e.g., satellite backup).

In summary, the Digital Twin evolves into an **executable knowledge system**, where the ***TBox*** governs meaning and rules, and ***ABox***/***RBox*** capture state, validate consistency, and enable inference, transforming data into actionable decisions.

5 Use Cases

5.1 Use Case I: Semantic Jurisdiction and Logical Governance in DT-II

Semantic Jurisdiction is the governance mechanism of the **DT-II** that establishes authority and policy compliance over digital entities based on meaning, context, and relations—transcending traditional geophysical boundaries. In this model, jurisdiction is no longer a static hardware attribute but a **normative entity mediated by Relators**.

5.1.1 Foundation and Axiomatization

Defined in the ***TBox*** through restriction axioms, Semantic Jurisdiction leverages the Reasoner to determine which norms (e.g., LGPD, GDPR, SLAs) apply to real-time data flows. In the ***RBox***, property chains formalize propagation: if a `NetworkNode` is bound to a legal norm, all `Channels` and `Connections` mediated by Relators inherit the same restrictions.

5.1.2 Role of the Digital Twin

The Digital Twin acts as custodian of jurisdiction. By ingesting telemetry (gNMI, YANG-Push, SNMP), the system populates the ***ABox***. Through **Semantic Elevation**, it verifies whether, for example, a health data packet traverses a node under a jurisdiction that fails to meet privacy requirements defined in the ***TBox***.

5.1.3 Operationalization and Resilience

In dynamic events (e.g., DDoS attacks, BGP failures), Semantic Jurisdiction guides the **MAPE-K** cycle Murch [2004]: (a) **Analysis**: Reasoner detects violations of jurisdiction axioms; (b) **Planning**: System searches for compliant alternatives; (c) **Execution**: Configuration is applied via ***NETCONF*** only after logical validation.

In summary, Semantic Jurisdiction transforms infrastructure into a ***legally aware*** system, where compliance is mathematically verified by DL, ensuring governance integrity in complex and mutable networks.

5.2 Use Case II: Behavior of the Model under a DDoS Attack

In a Distributed Denial of Service (DDoS) attack, response time is critical. The Digital Twin acts as a Semantic Immune System, leveraging the triad of ***TBox***, ***RBox***, and ***ABox*** within the ***MAPE-K*** cycle as follows:

- **Monitoring** Telemetry (gNMI, YANG-Push, SNMP) updates the ***ABox*** in real time. For example, a flow spike from 1 Gbps to 100 Gbps is recorded as:

```
inst:Unusual_Flow a ii:AnomalousFlow
```

- **Analysis** The Reasoner applies ***RBox*** property chains: (a) the anomalous flow saturates `Channel_Alpha`; (b) transitivity propagates risk to dependent services (e.g., DNS); (c) the origin is traced to a specific **Semantic Jurisdiction**.
- **Planning** Consulting the ***TBox***, the system applies rules: if a critical service is at risk and the origin is external, mitigation (Blackholing or Rate Limiting) must occur at the edge. The plan prioritizes nodes sustaining the network's **Logical Constitution**.
- **Execution** Logical decisions are translated into technical commands. Via ***NETCONF***, edge routers are reconfigured to filter malicious traffic.
- **Advantages of the Semantic Approach** Unlike conventional defenses that block IPs, the **DT-II** enables: (a) **Context Awareness**: cooperation agreements mapped in the ontology may trigger governance APIs instead of simple blocking; (b) **Mereological Resilience**: the Reasoner may sacrifice non-vital channels to preserve critical services; (c) **Traceability**: each mitigation step is explainable, linked to axioms in the ***TBox***; (d) **Compliance**: no action is executed without validation by the ***Logical Constitution***.
- **Sequence Flow** The mitigation occurs within milliseconds, moving from the **Active Infrastructure** to the **Logical Core** and back. Figure 2 illustrates the process.

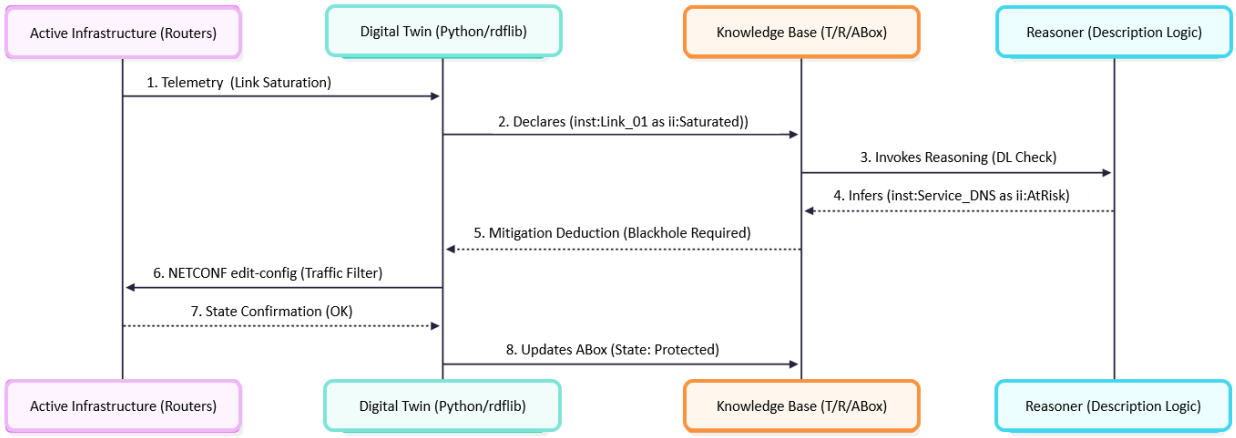

Figure 2: Sequence Diagram of DDoS Mitigation via DT-II
Sequence Diagram of DDoS Mitigation via DT-II

In summary, the ontology-based approach enables **policy-driven mitigation**, where assets of highest semantic and contractual value are prioritized, ensuring resilience and governance integrity.

6 Why Internet Infrastructure Requires a Project Like DT-II

The ***Internet Infrastructure*** (II) must be understood not merely as cables and protocols, but as a **living system** that has surpassed the limits of manual management. The **DT-II** project is necessary for four reasons: (R1) **From Static Management to Semantic Autonomy**: Current II management relies on rigid scripts and manual CLI commands. DT-II introduces **Logical Reasoning**, enabling the system to deduce solutions for complex failures and attacks under a ***Logical Constitution***; (R2) **NETCONF as Interoperability Catalyst**: The growing adoption of ***NETCONF*** provides a standardized, vendor-agnostic channel for the Digital Twin to interact with the **Active Infrastructure**, updating ***ABox*** instances in real time; (R3) **Global Semantic Governance**: In a fragmented regulatory landscape (e.g., LGPD, GDPR), DT-II codifies laws and SLAs as axioms, ensuring compliance regardless of geographic routing complexity; (R4) **Universal Semantic Interoperability**: Grounded in ***UO*** (***UFO***), DT-II resolves data silos, acting as a universal translator across domains (5G, Satellite, SD-WAN[2]), reducing costs and integration errors.

In summary, without DT-II, II remains an **Active Infrastructure**—processing packets without context. With DT-II, II gains **State Awareness**, evolving into an **Autonomic Network** that understands what it moves, why it moves, and whether it has logical permission to do so.

[2]Software-Defined Wide Area Network, a software-based approach to WAN management.

7 Conclusion and Future Works

This paper introduced the conceptual foundations of the **DT-II**, a Digital Twin for Internet Infrastructure grounded in Ontology, Knowledge Graphs, and Description Logic. By formalizing governance through **Semantic Jurisdiction** and enabling policy-driven responses to complex events such as DDoS attacks, the model advances the vision of transforming II from a passive infrastructure into an **autonomic, legally aware system**. The proposed architecture demonstrates how ontological rigor can ensure resilience, compliance, and semantic interoperability across heterogeneous domains. All results from this project will be made available at https://github.com/LJuliaoBraga/dtii/tree/main.

Future work will focus on: (i) simulation of the Digital Twin to evaluate alarms in Autonomous Systems (AS); (ii) experimental validation of spatiotemporal response mechanisms; (iii) integration with programmable infrastructures via ***NETCONF***; (iv) exploration of semantic interoperability across domains such as 5G, satellite, and SD-WAN; (v) to participate in the evaluation of the two proposed telemetry alternatives—YANG-Push and gNMI—while also remaining open to those that may emerge in the near future; and (vi) Integration with Quantum Key Distribution (QKD) and Quantum Internet: We plan to explore the use of Quantum Communication channels for the transmission of highly sensitive telemetry and governance metadata between the Active Infrastructure and the Digital Twin. As the DT-II handles critical "Semantic Jurisdictions" involving state-level privacy and strategic infrastructure topology, the integration of QKD will ensure that the synchronization between the physical and logical layers remains secure even against future quantum computing adversarial threats, maintaining the "Logical Constitution's" integrity through a provably secure quantum-classical hybrid architecture Kozlowski et al. [2023], Wang et al. [2024], Meter [2014]. These steps aim to evolve the model from conceptual design to experimental deployment, culminating in its 2026 submission as a draft to the IRTF ***nmrg*** within the CGI-IETF initiative, in collaboration with the following Brazilian universities: **UFABC** (SP), **UFRGS** (RS), **IFSPe** (PE), and **UNICAP** (PE).

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